

● MEDIUM

● **ADVANCED MATH**

Advanced Math

04

10 questions · ~15 min

Q1

GRID-IN ADVANCED MATH

The exponential function g is defined by $g(x) = 19 \cdot a^x$, where a is a positive constant. If $g(3) = 2,375$, what is the value of $g(4)$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$b - 72 = \frac{x}{y}$$

The given equation relates the positive numbers b , x , and y . Which equation correctly expresses x in terms of b and y ?

A. $x = \frac{b - 72}{y}$

B. $x = by - 72$

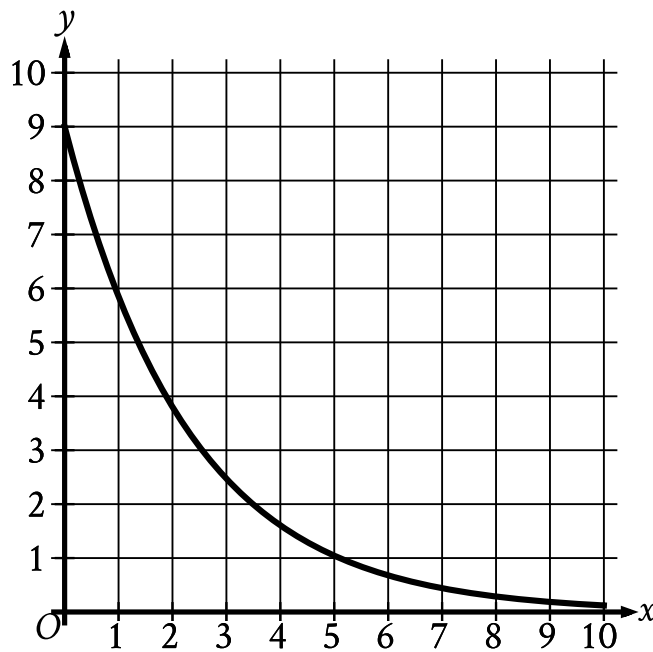
C. $x = \frac{by - 72}{y}$

D. $x = by - 72y$

$$(x - 11y)(2x - 3y) - 12y(-2x + 3y)$$

Which of the following is equivalent to the expression above?

- A. $x - 23y$
- B. $2x^2 - xy - 3y^2$
- C. $2x^2 + 24xy + 36y^2$
- D. $2x^2 - 49xy + 69y^2$



- Moving from left to right:
 - The curve is in quadrant 1.
 - From point (0 comma 9), the curve trends down sharply.
 - From point (5 comma 1), the curve trends down gradually.

The graph gives the estimated number of catalogs y , in thousands, a company sent to its customers at the end of each year, where x represents the number of years since the end of 1992, where $0 \leq x \leq 10$. Which statement is the best interpretation of the y -intercept in this context?

- A. The estimated total number of catalogs the company sent to its customers during the first 10 years was 9,000.
- B. The estimated total number of catalogs the company sent to its customers from the end of 1992 to the end of 2002 was 90.

- C. The estimated number of catalogs the company sent to its customers at the end of 1992 was 9.
- D. The estimated number of catalogs the company sent to its customers at the end of 1992 was 9,000.

$$v^2 = \frac{LT}{m}$$

The formula above expresses the square of the speed v of a wave moving along a string in terms of tension T , mass m , and length L of the string. What is T in terms of m , v , and L ?

A. $T = \frac{mv^2}{L}$

B. $T = \frac{m}{v^2L}$

C. $T = \frac{mL}{v^2}$

D. $T = \frac{L}{mv^2}$

$$(2x+3)-(x-7)$$

Which of the following is equivalent to the given expression?

- A. $x - 4$
- B. $3x - 4$
- C. $x + 10$
- D. $2x^2 + 21$

x	fx
-1	10
0	14
1	20

For the quadratic function f , the table shows three values of x and their corresponding values of fx . Which equation defines f ?

- A. $fx = 3x^2 + 3x + 14$
- B. $fx = 5x^2 + x + 14$
- C. $fx = 9x^2 - x + 14$
- D. $fx = x^2 + 5x + 14$

$$y = 4x$$

$$y = x^2 - 12$$

A solution to the given system of equations is x, y , where $x > 0$. What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$x^2 + 6x + 4$$

Which of the following is equivalent to the expression above?

- A. $(x + 3)^2 + 5$
- B. $(x + 3)^2 - 5$
- C. $(x - 3)^2 + 5$
- D. $(x - 3)^2 - 5$

The area A , in square centimeters, of a rectangular cutting board can be represented by the expression $ww + 9$, where w is the width, in centimeters, of the cutting board. Which expression represents the length, in centimeters, of the cutting board?

- A. $ww + 9$
- B. w
- C. 9
- D. $w + 9$